# Social Curiosity: Study 4 (Uncertainty) (#229436)

**Authors**
This pre-registration is currently anonymous to enable blind peer-review.
It has 2 authors.

**Pre-registered on:**
2025/05/20 16:24 (PT)

**1) Have any data been collected for this study already?**
No, no data have been collected for this study yet.

**2) What's the main question being asked or hypothesis being tested in this study?**
We hypothesize that 4- and 5-year-old children would be more curious to find out how others evaluated their own work when they are more uncertain about whether they received a positive evaluation. Specifically, we predict that children would be more likely to look at their own evaluation (rather than another individual's) when they completed a novel, difficult task than when they completed a simple task they had done before.

**3) Describe the key dependent variable(s) specifying how they will be measured.**
After an experimenter evaluates the participant's tracing (either a Line or a Bird, see "Conditions") and another child's (i.e., "Jordan", an unfamiliar, gender-matched child to the participant) tracing out of sight, they will give children permission to look at one of the evaluations. The key dependent measure is which folder children choose to look inside of.

**4) How many and which conditions will participants be assigned to?**
Children will be randomly assigned to the Line-Line-Line condition, or the Line-Line-Bird condition. First, children across both conditions will receive two rounds of evaluation from the experimenter on two identical line tracings, each time receiving positive evaluation in the form of a star. Then participants will be presented with a final tracing: either the same line tracing (Line-Line-Line condition), or a bird tracing (Line-Line-Bird).

After children complete their final tracing, they will be told that another unfamiliar child made a tracing. The two tracings will be placed inside of separate folders and evaluated out of sight.

**5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.**
We will run the following Bayesian generalized linear model in the rstanarm (Goodrich et al., 2024) package in order to determine whether folder choice differs by condition:

stan_glm(formula = choice ~ condition,
family = binomial,
data = data)

"Choice" corresponds to the primary dependent measure (i.e., whether children choose their own folder (coded as 1) or Jordan's folder (coded as 0)), and "condition" corresponds to either the Line-Line-Bird condition (coded as 1) or Line-Line-Line condition (coded as 0; between-subjects).

We will use the package default of weakly informative priors (normal distributions on coefficients with SD = 2.5, scaled to predictor magnitudes).

In order to assess the hypothesis of an effect of condition on choice, we will compute the Bayes Factor using the bridgesampling (Gronau et al., 2020) package in R. We will compare a model with a condition term (i.e., choice ~ condition) and one without (i.e., intercept-only model; choice ~ 1;):

bf(conditionterm_model, interceptonly_model)

We will interpret a BF10 > 10 as strong evidence, > 5 as moderate evidence, and > 3 as weak but meaningful evidence for a condition effect. We will use the same thresholds to interpret evidence in favor of the null (i.e., BF01)

**6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.**
Participants will be excluded for interference (teachers, children, or other individuals), experimenter error (e.g., large deviations from the study script), failure to complete the study, looking inside one of the folders prior to the key test phase, ambiguous or uncode-able choice, having parent-reported developmental delays, significant distraction away from the study, or technical errors (e.g., video does not record).

Finally, participants will be asked comprehension questions throughout the study (a total of four times). In each instance, participants will first be asked a free response question (e.g., "Do you remember what [the star slip] means?"); if they fail this question, they will then be asked a forced-choice question (e.g., "Do you think it means I thought your tracing was really good? Or just okay?"). If participants do not answer the free response question correctly at least once, their data will be excluded from analysis.

**7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.**
We are uncertain as to the size of the effect we will observe (if any), thus we plan a sequential Bayes Factor analysis (e.g., Mani et al., 2021). We will

calculate the Bayes Factor via model comparison (see "Analyses"): comparing a model with a condition term (i.e., choice ~ condition) and one without (i.e., intercept-only model; choice ~ 1).

We will test an initial sample of 20 children (4- and 5-year-olds) in each condition, then evaluate the Bayes Factor on the hypothesis of an effect of condition on folder choice (i.e., bf(conditionterm_model, interceptonly_model)) after every 4 data points (2 per condition). We will stop testing at BF10 > 10, BF10 < ⅕, or at N = 72 (36/condition).

**8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)**

For consistency with previous studies in this line of work, we will also conduct a frequentist analysis of the data (Fisher's Exact Test) at the conclusion of data collection.